MODULE 10

Cryptocurrency Crime Investigation

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What You Will Learn

Bitcoin is not untraceable - it's pseudonymous. Every transaction is public forever. That's your weapon.

- **Bitcoin addresses** - validity, balance, history, first-seen, on a free explorer
- **Transaction anatomy** - inputs, outputs, fee, change addresses
- **Chain analysis** - hop-by-hop tracing, clustering, the common-input rule
- **The exchange** - the criminal's weak link, and BNSS s.94 requests that work

What You Will Learn (2)

Plus the tricks criminals use - and the mistakes investigators must avoid.

- **Mixers and privacy coins** - red flags, scoring, and what changes
- **Stablecoins and new trends** - USDT networks, DeFi fraud, rug pulls
- **Victim interviews** - the 10 questions that build half your case file
- **Real cases** - Bitfinex $3.6B, and an Indian investment-app gang

The Inspector Was Wrong

"Money gone in Bitcoin will never come back. It's untraceable." - a confident inspector. He was wrong.

- Bitcoin is **PSEUDONYMOUS**: no names appear, but EVERY transaction stays public
- The blockchain **never lies and never forgets** - even years later
- This module teaches you to **follow the money yourself**

Blockchain: A Ledger Open to Everyone

A bank's ledger is locked inside the bank. The blockchain's ledger lies open on the internet:

- Every transaction sits in a block; blocks chain together - written once, **never changed**
- Thousands of computers hold copies - **no one can erase the record**
- Anyone can see: which address sent how much to which - **no notice needed**

Bitcoin Address Formats

Every investigation starts with an address. First, read it correctly:

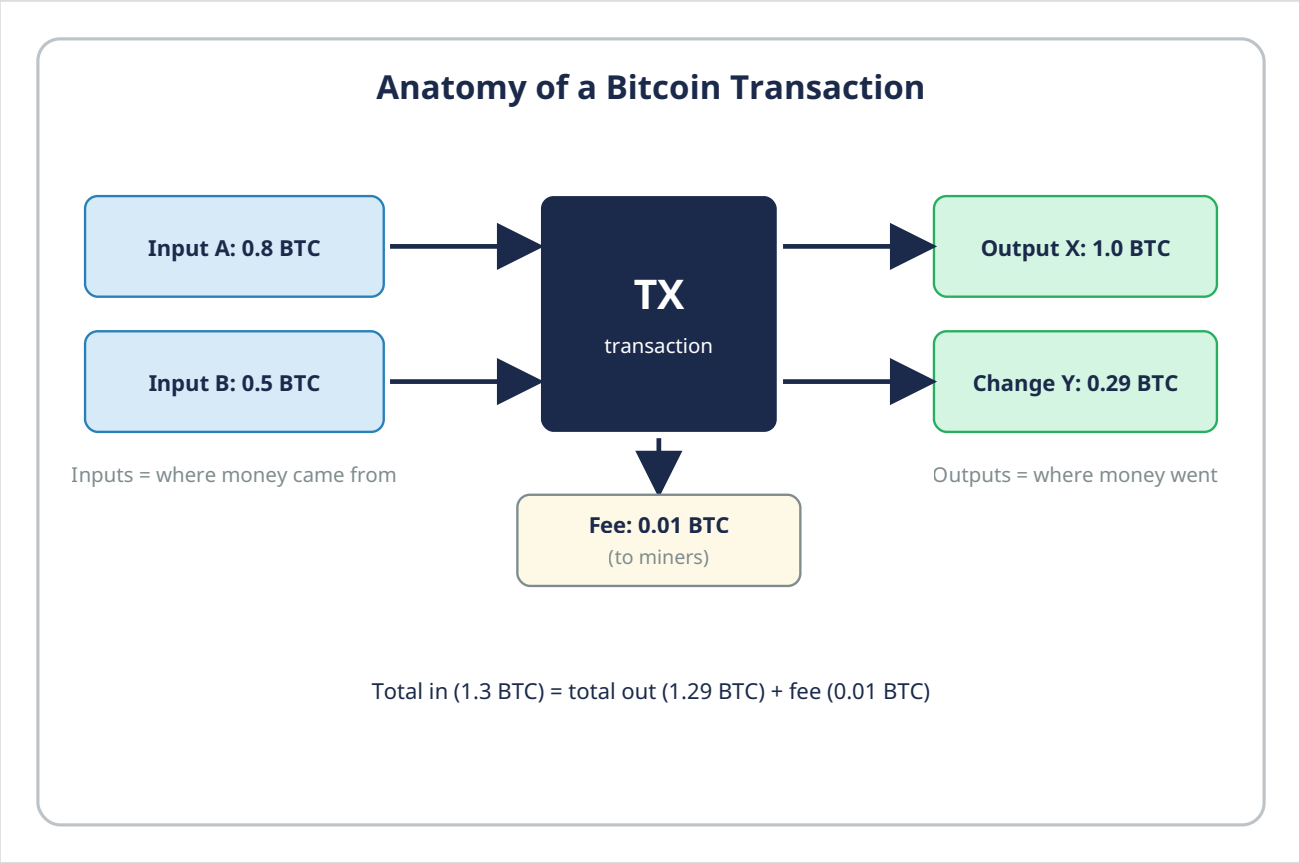
Format	Starts with	Length
Legacy	1...	26-35 characters
P2SH	3...	26-35 characters
Bech32	bc1...	42-62 characters

What One Address Tells You

One address gives you four facts - all public, no permission needed:

- **VALIDITY:** the checksum catches wrong addresses - money sent wrong is lost forever
- **BALANCE + totals:** high received but zero balance = the money already moved on
- **TRANSACTION HISTORY:** every move in and out, with dates
- **FIRST SEEN:** new address, 1-2 transactions = created for ONE crime

Anatomy of a Transaction



- **INPUTS:** where money came from (can be many) | **OUTPUTS:** where it went
- **Fee = total inputs MINUS total outputs** - always verify the math
- **CHANGE address:** unspent money returns to the sender's new address

Reading a Transaction on the Explorer

Open any transaction on blockchain.com. Read it in this order:

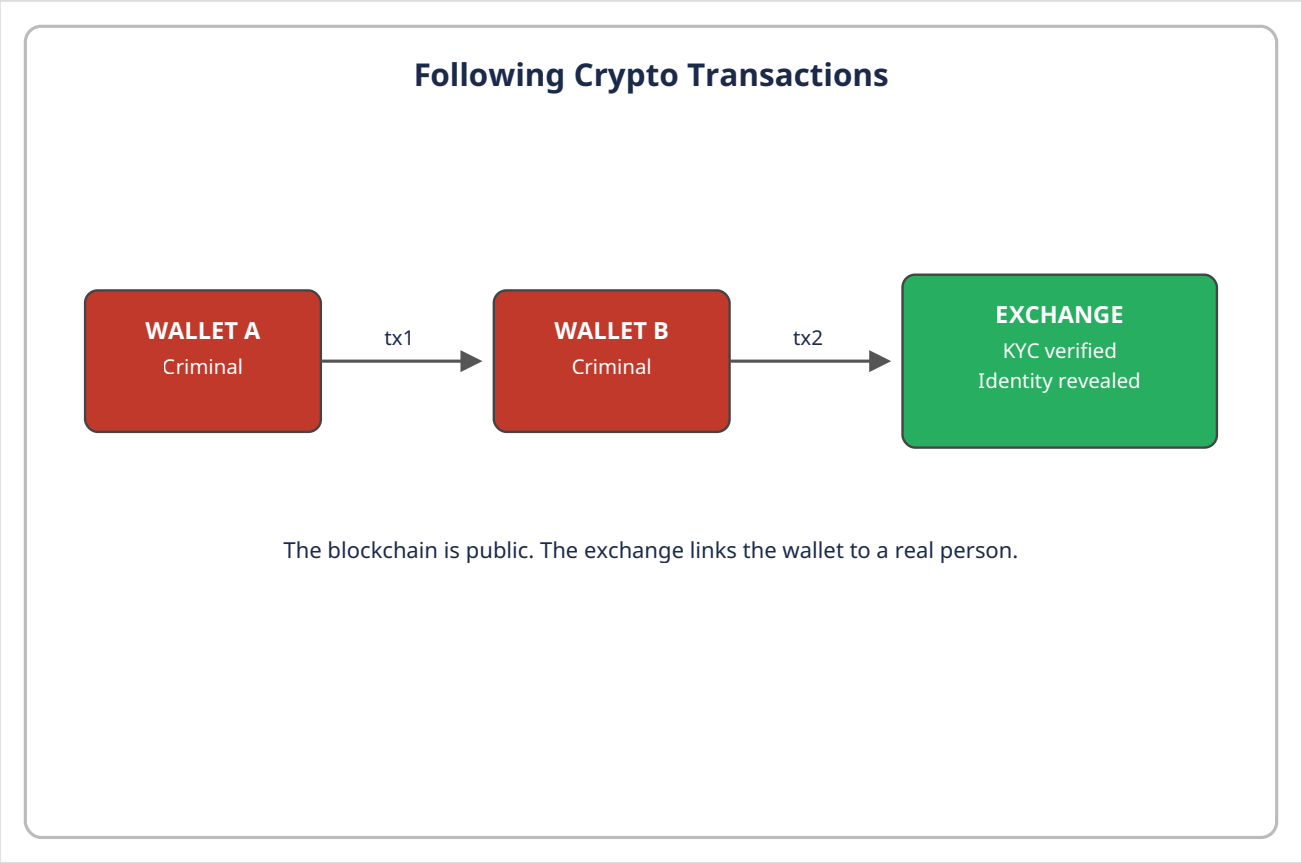
- **Inputs first** (source), then outputs (destination) - add both, the difference is the fee
- **Confirmations:** 6+ means the transaction is final
- Time shows in **UTC** - convert to IST (+5:30) for the case diary, note both
- The **largest output** usually carries the money forward - follow it

Chain Analysis: Joining the Links

The fraudster rotates money through addresses, thinking it hides him. Every hop is recorded:

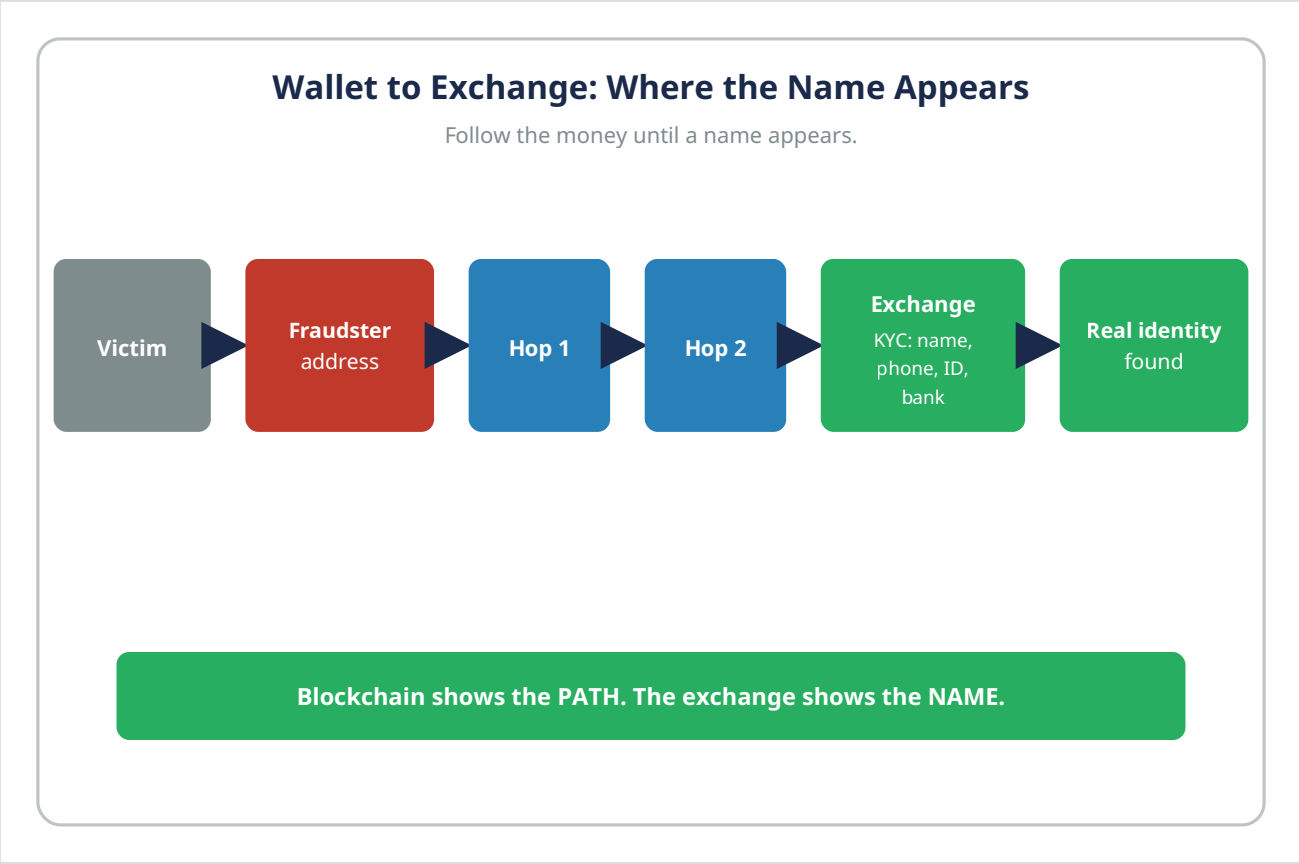
- Follow **hop-by-hop**: largest output → next transaction → repeat 3-4 hops
- **COMMON-INPUT RULE:** 3 addresses funding one transaction = same wallet, same person
- **PEEL CHAIN:** big amount, small pieces peeled to new addresses - clearly visible
- Tracing gives the PATH, not the name - **the name comes from the exchange**

Tracing Hop-by-Hop



- Hop 0: victim's payment → Hops 1-4: follow the largest outputs
- At each hop note: date-time, amount, from, to, special signs
- Trail hits a known exchange address? **STOP tracing, START legal process**

The Exchange: The Criminal's Weak Link



- Criminal must cash out: crypto → exchange → rupees in bank
- Exchanges hold **KYC:** name, ID proof, phone, bank account
- Your BNSS s.94 notice turns a wallet address into a real person

The 5-Point Exchange Notice

A hasty, incomplete notice gets no reply. Your notice MUST carry these five:

- **Complete Bitcoin address** + the transaction hash in question
- **Date and time** - in IST, with UTC alongside
- Exactly what you need: KYC, login IPs, linked bank account, transaction log
- **Case reference:** FIR number, sections, IO's name and contact

Speed and Preservation

Indian exchange = fast reply. Foreign exchange = MLAT, and MLAT takes months:

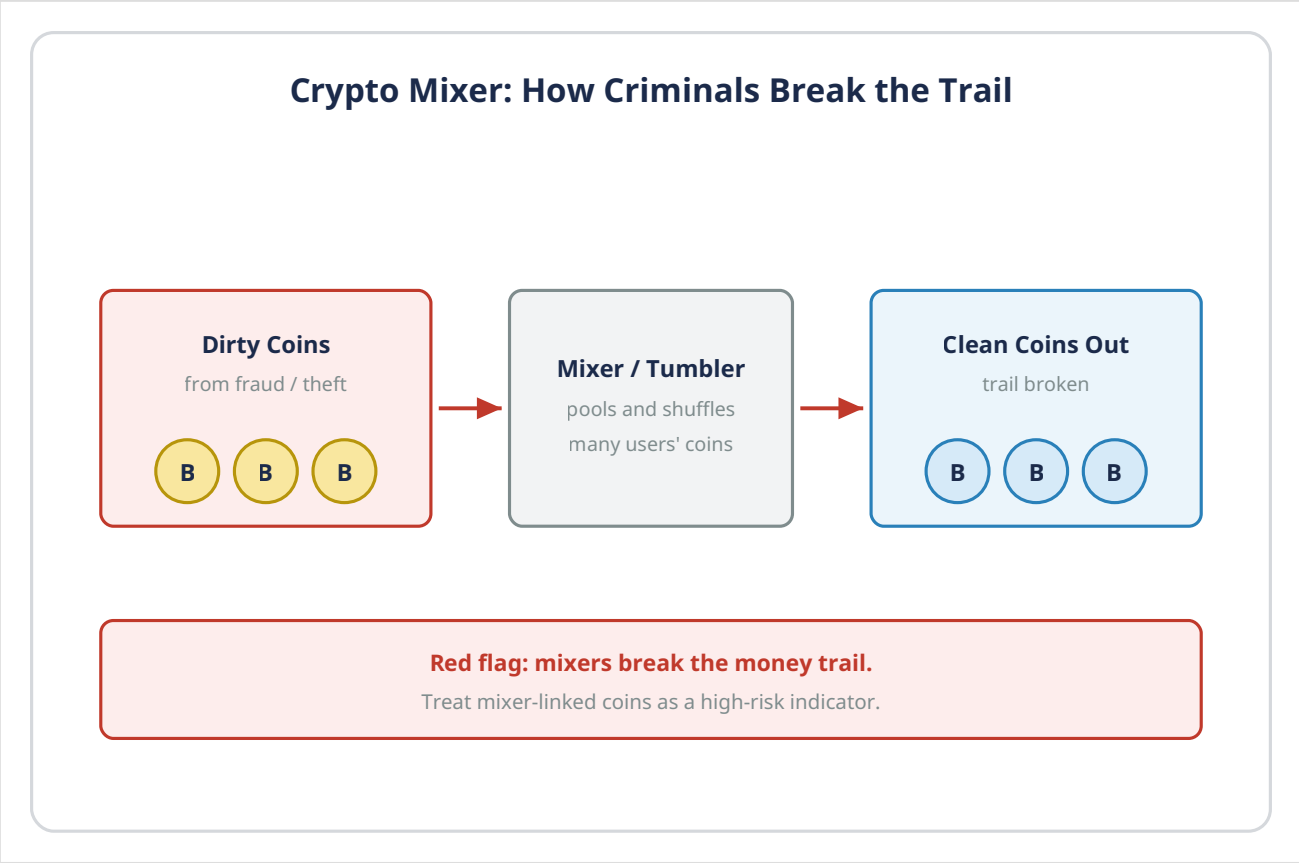
- Send the notice on **DAY ONE** - don't wait for tracing to finish
- Also ask the exchange to **PRESERVE the data** - records get deleted while you wait
- Secure the victim's own exchange records too - his purchase is also evidence
- Crypto moves in minutes; **your paperwork must move in hours**

Wallets: Hot and Cold

A wallet is a digital purse. Two types - and seizure rules that decide everything:

- **HOT wallet:** on the internet (app, exchange account) - traceable
- **COLD wallet:** offline (hardware device, paper) - hard to catch, opens on seizure
- **SEIZURE:** airplane mode / Faraday bag ON, keep device POWERED ON, photograph the screen NOW
- 12-24 English words on paper = **seed phrase** = the money itself

Mixers: The Criminal's Fog



- Mixer pools many people's coins and redistributes - breaks the visible link
- **Tornado Cash:** sanctioned by the US Treasury - its name in a trail is a giant red flag
- Criminals still err BEFORE or AFTER the mixer - KYC, email, phone

Mixer Red Flags: Score It



- Round amounts (1.0 BTC), rapid hops, peel chains, all-new addresses
- Score: 0-1 low, 2-3 medium, 4+ high suspicion - always write **"suspected"**
- Pattern is suspicion, NOT proof - proof needs KYC + legal process

Privacy Coins and Chain Hopping

When Bitcoin tracing gets hot, criminals switch tools:

- **MONERO:** hides sender, receiver AND amount - harder than Bitcoin
- **CHAIN HOPPING:** Bitcoin → Monero → Bitcoin across exchanges - path breaks at each hop
- But every hop touches an exchange - **and exchanges keep records**
- That's why exchange data beats any tracing trick

Stablecoins and New Trends

Criminals love stable prices. Know where the new money moves:

- **USDT on TRON (TRC20):** low fees, criminal favourite - use tronscan
- **USDT on Ethereum (ERC20):** use etherscan - wrong network = you find NOTHING
- New frauds: **DeFi schemes, fake NFTs, RUG PULLS** (new coin, collect money, vanish)
- Trends change; the method doesn't: chain path + exchange identity

5 Red Flags of Crypto Fraud

Teach these five in every awareness session - with a story, not dry rules:

- **"Double your money in a month"** - no real investment works like that
- Hurry pressure: **"invest TODAY, offer ends tomorrow"**
- Unknown app pushed instead of a reputed exchange
- **PIG BUTCHERING:** friendship first, investment advice later

The Withdrawal Trap

The fifth red flag deserves its own slide - it's where victims finally realize:

- After investing, **withdrawal fails**: "deposit tax first, then it will come out"
- Each "fee" is another theft - the money was never invested anywhere
- If all five signs appear together, **fraud is happening** - act, don't debate
- Awareness is the first defence - **stories stick, rules don't**

Questioning the Victim (1-5)

The victim is panicked. Ask calmly - these answers are half your case file:

#	Question
1	Which crypto did the fraudster ask for?
2	Where did he send the address - WhatsApp, Telegram?
3	Which exchange or app did YOU buy from?
4	Exact date and time of sending the money?
5	Amount - in crypto AND in rupees?

Questioning the Victim (6-10)

Second half - the technical details that start the trace:

#	Question
6	Transaction hash or reference number?
7	The fraudster's address - screenshot available?
8	Any website or app he made you download?
9	Did you share any OTP or screen-share?
10	Last contact - when, from which number or ID?

5 Investigator Mistakes

Five mistakes that kill crypto cases before they start:

- **"Crypto is untraceable"** - the myth that ends investigation on day one
- **Delay:** money rotates through wallets in MINUTES - preserve exchange data immediately
- **Pattern = proof:** a mixer-like pattern is suspicion only, never an accusation
- Entering a wallet or account yourself - explorers are legal, **entry without authority is a crime**

The Case File on Paper

Tracing done is half the victory. Recording it right is the other half:

- Screenshots must show **URL, date and time** - never a cropped fragment
- Export explorer data and note its **hash** - kills "data was changed" claims
- Write every step's date in the case diary - the digital chain of custody
- Lab outputs (trail table, checklist) go in the file **with dates**

Professional Tools: Level Up

Free explorers start the trail. Big cases need professional tools:

- **CHAINALYSIS / ELLIPTIC / CIPHER-TRACE:** label addresses - exchange? mixer? market?
- **Risk scores:** how deep an address's crime connection runs
- **Alerts:** money moves from a flagged address - you know instantly
- Your trail table feeds these tools - **extract the lead free, confirm it professional**

Real Case: Bitfinex - $3.6 Billion

2016: a massive theft from the Bitfinex exchange. The coins rotated for YEARS:

- Criminals split them into small pieces, hop after hop
- 2022: US agencies traced the **full path** through chain analysis
- **$3.6 BILLION** in Bitcoin seized - the blockchain never forgot
- Lesson: even a years-old case stays alive. **Never give up.**
- Source U.S. Department of Justice

Real Case: Fake Investment App Gang

An Indian gang ran a fake crypto investment app - high returns, vanishing money:

- Money traced **hop-by-hop** on the blockchain to an exchange
- KYC from the exchange named the accused
- Same method works in India: address → chain path → exchange identity
- Lesson: the ED walks this exact path
- Source Enforcement Directorate, via PIB

Your Turn: Labs in This Module

Four labs - public explorers only. No wallet connections, no money moved:

- **Lab 1:** read a real address's full history - balance, first-seen, fees
- **Lab 2:** identify address formats - Legacy, P2SH, Bech32
- **Lab 3:** trace 3-4 hops, build the trail table, draw the money diagram
- **Lab 4:** score the trail on the mixer red-flag checklist

Key Takeaways

If you remember nothing else from Module 10, remember this:

- The blockchain is an open ledger - **start from the address**
- Fee = inputs minus outputs; change addresses belong to the sender
- Common-input rule and peel chains reveal hidden wallets
- The exchange is the weak link - **KYC turns addresses into names**
- Pattern is suspicion; proof needs legal process
- **Speed wins:** preserve exchange data in the first 48 hours

Review Questions

- 1. An address received 50 BTC total but holds 0 now, first seen 3 days ago, 2 transactions. What does this tell you? What's next?
- 2. At hop 3, money splits into 40 tiny outputs to new addresses. Name the pattern - and explain why you still can't arrest anyone.
- 3. Draft the five points of a BNSS s.94 notice to an exchange. Why include both IST and UTC?
- 4. The trail enters Tornado Cash. What changes in your investigation strategy from this point?
- 5. A victim's USDT went to a TRC20 address. You search it on etherscan and find nothing. What went wrong?

NEXT: MODULE 11

Social Engineering Crimes and Attacks

Manipulation tactics & victim psychology

Cyber Crime Investigator - Training Manual